You should have noticed that the pattern used contains no quantifiers for the literal characters (though the recursion is quantified with ?), yet we're able to match the full string 'aaabbb'. If the pattern had simply been /ab/ our match would have been just the two middle characters from the sample, by adding (?R)? we're asking the PCRE engine to keep matching a recursively, zero or one times and the zero is important because it allows the recursion to exit. Once the recursion is done, the rest of the pattern (a single 'b' character) is matched recursively. We have three 'a' characters, which means three levels of recursion, so there need to be three 'b' characters since they will be matched recursively too. The pattern /a(?R)?b/ will match any string that has any amount of 'a' followed by an equal number of 'b'.

PRCE Functions

Now we'll turn our attention to the functions that PHP provides for using regular expressions. The full list of core regex string functions is as follows:

Table 8-35. *PREG functions*

preg_filter()	Search sample with regex and replace matching portions
preg_grep()	Search array of strings and return matching elements
preg_last_error()	Get error code from the most recent PCRE call
preg_last_error_message()	Get error message from the most recent PCRE call
preg_match()	Search sample, return on first match
preg_match_all()	As preg_match(), but with implicit global modifier – returns all matches
preg_quote()	Escape all PCRE meta-characters in a string with backslash
preg_replace()	Search and replace matches with supplied string
preg_replace_callback()	As preg_replace() but using callable for replacement
preg_replace_callback_array()	As preg_replace_callback(), mapping patterns to callables
preg_split()	Split a single string into array of sub-strings, using a pattern